Communities of the Interested

Redistricting Institution Congruence with Communities of Interest Among Geocoded Public Comments

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1 Introduction

1.1 Redistricting and Congruence

Existing literature has found that state legislatures are generally responsive to public opinion but not congruent with the concrete policies desired by public majorities (Lax and Phillips 2012). Like many other policy issue areas, the clandestine practice of redistricting is not immune to this congruency gap. Partisan legislature redistricting outcomes often produce large "efficiency gaps" between statewide vote shares and resulting seat counts for each political party (Warshaw, McGhee, and Migurski 2022). Redistricting, especially in recent cycles, has also widely reduced the frequency of both competitive electoral districts (Jasny et al. 2025; Kenny et al. 2023) and majority-minority or majority-influence districts (Barabas and Jerit 2004). However, unlike most issue areas, redistricting decisions produce categorical and geographically-defined results on the same dimensional plane as any hypothetical, concrete public preference about redistricting. Instead of trying to evaluate how the specific language of a piece of legislation translates onto individual policy preferences, which can be untethered by practical and semantic constraints, redistricting decisions present an opportunity for both policy and mass opinion can be mapped onto sets of precincts in a given state with measurable degrees of overlap. This makes redistricting a powerful arena for the study of congruence more broadly, especially with redistricting being so closely linked to the health of America's state and federal democracies (Grumbach 2023; Gelman and King 1994; Fraga 2016).

Measuring congruence with redistricting decisions, however, requires that we understand what most members of the public would request of their boundary-drawing institutions in the first place. This is no easy task. Public knowledge and opinions on redistricting, as a concept, are relatively sparse (Cooper et al. 2024; McCarthy 2022), with respondents often relying on partisan cues or judgements about the overall redistricting process. Evaluating public opinion on redistricting is further complicated by low levels of geographic literacy across the country (McFarlane 2024), meaning that citizens are unlikely to have the expertise to properly judge an entire state's district map. True opinions on a map among the public are likely localized and related to the partisan or nonpartisan factors of the communities Americans call home. A citizen might not have a good sense of how to divide up an entire metro area into state house districts,

but they can probably comment on whether their neighborhood, city, school district, and county should be kept whole, combined with other areas, or separated from other areas. Place based identities are a powerful force in the United States (Elder and Lueders 2025), implying that residents of a community have some perception of the social, cultural, economic, and geographic dynamics that separate "where they live" from everywhere else. It is under these conditions that a key redistricting criteria - the concept of "Communities of Interest" - can offer a critical advantage over other outcome heuristics in the translation of public opinion to geographic configurations that enacted maps can either be congruent or incongruent with.

1.2 Communities of Interest in Legislative Redistricting

Nicholas Stephanopoulos (2012) articulates that a group of people constitute a community of interest (COI) if they (1) share a common geographic range, (2) share common socioeconomic or cultural interests, and (3) subjectively identify as members of a cohesive geographic community. Respecting communities of interest is typically one of several criteria that many boundary-drawing institutions are constitutionally obligated to respect (Webster 2013), including contiguity, compactness, population equality, and protection of minority voting populations. However, these criteria are difficult to balance, and some can even be contradictory. In fact, the "reapportionment revolution" of the 1960s, a series of Supreme Court decisions requiring that legislative redistricting strictly adhere to the criteria of population equality across districts, is credited with massively disrupting the ability of states to respect COIs despite the clear benefits of meeting both criteria simultaneously (Stephanopoulos 2012). State and federal courts are nonetheless receptive of arguments in favor of protecting COIs, often striking down maps in clear violation of the criteria. Furthermore, recent court decisions have diminished the prominence of the Voting Rights Act as a basis for striking down statewide maps that fail to appropriately represent minority populations (Chen et al. 2022), leaving communities of interest as a possible line of defense for these populations to still litigate with.

Thus, a burgeoning community of scholars, recognizing the utility offered by communities of interest as well as their importance in district plan litigation, have articulated a variety of methods for COI measurement. Some scholars have used contextual and behavioral characteristics of smaller geographic units within states (Kruse, Gao, and Mayer 2026; Gimpel and Harbridge-Yong 2020; Makse 2012; Syme 2025; Rossiter,

Wong, and Delamater 2018; Swanson and Bryan 2024), clustering units into cohesive groups and comparing district maps to group boundaries. For example, Kruse, Gao, and Mayer (2026) use Facebook friendship networks and mobile device visits between Census block groups to construct COIs, while Maske sorts communities based on issue area initiative votes in Ohio (Makse 2012). These observational approaches present defensible solutions based on well-distributed data across entire states, and their sophisticated methods have yielded actionable insights, but they nonetheless ascribe community membership at a distance from the public's participation in the redistricting process.

Other scholars, therefore, have constructed COIs based on grassroots input from the public, including the deployment of public mapping tools that allow respondents to "draw" their communities of interest from a bird's eye view (Chen et al. 2022; Chambers et al. 2022). These drawn communities of interest can then be compiled and analyzed using a variety of computational techniques. Chambers and colleagues cluster overlapping COIs based on their similarity across respondents, while Chen and colleagues introduce novel measures to evaluate how egregiously an enacted plan splits a given community. These studies provide a framework that boundary-drawing institutions can use to engage the public and better link redistricting with democratic participation. They also provide a closer estimation of the kind of communication flows between constituents and legislators that are likely to occur during the redistricting process.

The most widespread attempts at measuring communities of interest, however, have thankfully been made the states themselves, with boundary-drawing institutions launching widespread outreach programs in recent redistricting cycles. Many institutions offer public hearings so that concerned citizens can deliver speeches and presentations to boundary drawers, and lengthy essay submissions or even complete district maps drawn using publicly available GIS software are common forms of participation. But the largest body of testimony collected in the most recent redistricting cycle came from state-run public web comment platforms, with submissions numbering in the hundreds or thousands for many states, especially those making use of an independent redistricting committee. Among these comments, communities of interest discussion often naturally takes center stage, assembling a rich mosaic of geographic configuration requests that span entire states. As such, these web comments provide the strongest link to date connecting politically engaged residents to the redistricting process, but questions remain regarding how well these comments represent the

public and whether any requests are congruent with enacted maps.

Not only are these recent testimony gathering efforts widespread, but they also reflect, as with many other issue areas, the data that legislators themselves collect to gather opinions from constituents and make subsequent decisions, making public web comments ideally situated for answering several questions about if and why congruence lags during the redistricting process. Are public commenters independent testifiers resembling a cross section of their communities, or do they form unrepresentative, coordinated networks of advocacy across space and time? Are redistricting institutions responsive to requests of a certain nature or popularity? Across states, what types of redistricting institutions are the most responsive to community of interest testimony? Answers to each of these questions can lend insight to the broader study of public testimony, responsiveness, congruence, and democratic representation.

1.3 Theories of Congruence with Public Comment

Gabriella Subia's exploration of public participation during the creation of Colorado's eighth congressional district suggests that the state's IRC prioritized the representation of Hispanic voters, a competitive partisan balance, and the protection of various industries north of Denver, all of which were advocated for by both interest groups and local residents (Subia 2024). This case study aligns itself with the broader literature on public comment during rulemaking, which has generally found that institutions are responsive to those who participate (Sahn 2025; Ingrams 2023). One source of district map incongruence, then, might derive from the fact that those who testify have their requests met, but they don't appropriately represent a state's constituency, leading to the following hypothesis:

> H1: Across individual states, public web comment density and commenter demographics will not be representative of local community sizes and contexts.

Another analysis by Olivia Rambo finds among a sample of comments submitted to the California Citizen's Redistricting Commission that certain interests were more likely to achieve their boundary drawing aims than others, and many comments were copies of templates distributed to the public by interest groups (Rambo 2021). This study is consistent with the idea that even among those who participate, the most

well-funded and well-coordinated voices during the process of collecting public testimony have the greatest influence (Yackee and Yackee 2006), highlighting a second source of disconnect between mass opinion and redistricting outcomes. These organized interests can recruit or inspire bundles of comments making similar requests from similar areas at similar times, either using templates (Rambo 2021) or encouraging commenters to express a request in their own words.

Furthermore, for an issue like redistricting, an issue that the public has a limited knowledge of (Cooper et al. 2024), interest groups can either advocate for policy that appropriately represents the interests of their corresponding communities, or they can try to pull a district plan farther away from the will of underrepresented communities. Other comments will appear organically among the public, with altruistic, independent commenters - whether they be active community members, academics, or political hobbyists - genuinely expressing the best interests of a local population. Under the relative anonymity of many web commenting platforms, a boundary-drawing institution might struggle to discern the motivations behind popular requests among the comments they receive. Furthermore, boundary-drawing institutions can defend their decisions against the threat of litigation or veto by claiming they're responsive to the input they received, arguing that the most outspoken voices during the redistricting process are well informed about the appropriate decisions for their communities more broadly. Thus, we test the following two hypotheses about the nature of public web comment corpora:

> H2: Across individual states, public web comments will not be independent, and commenters will span a network of advocacy detectable by clusters of spaciotemporal comment density.

> H3: Within a state, congruence with some subset of public comment requests will be higher than what would be expected in the absence of public comments.

Which requests, contingent on support of H2 and H3, should we expect to be more congruent than others? A single state's redistricting institution has a collective set of goals and a necessarily unique approach to synthesizing any public opinion they receive. Previous research has found that legislators will often disregard public input that they disagree with (Butler and Dynes 2016), and boundary drawers must also

balance a variety of other redistricting criteria besides adherence to communities of interest (Webster 2013). Thankfully, McCartan and colleagues have built measures based on a Nash equilibrium process to assess the extent that partisan actors have leeway over the creation of final maps - based on initial boundary drawing responsibilities, veto guardrails, and the likelihood of court challenges - from state to state (2026). Unsurprisingly, states with independent redistricting commissions tend to have the lowest "partisan leeway" scores, while states placing the responsibility solely in the hands of a partisan legislature have much higher scores. States with greater partisan leeway are likely to be choosier about the public community of interest comments they consider, adhering closely to clusters of requests that align with their broader agenda while completely discounting others. It may also be the case that systematic differences in congruence emerge based on the racial or geographic makeup of a request's commenters or the nature of the request itself. This leads to the following hypotheses:

> H4: The association between request congruence and request popularity will be less predictable among states with greater partisan leeway over the redistricting process.

> H5: The association between request congruence and compositional features of a request's commenters will be stronger among states with greater partisan leeway over the redistricting process.

Support for each of these hypotheses can inform how congruence is disconnected from public opinion across many other issue areas (Lax and Phillips 2012). Unlike these other issues, however, the geographic results of redistricting enable direct measurements of congruence with constituent requests, and the criteria of communities of interest serves as a vital link between credible mass opinion and actionable geographic configurations. Exploring the nature of testimony that institutions collect and respond to can inform how statehouse democracies, some more than others, unintentionally or deliberately misalign themselves with the will of the people.

2 Data and Methods

2.1 Public Comments in the 2021-2022 Redistricting Cycle (forthcoming)

2.2 Public Comment Data in Colorado

Public web comment testimony for the 2021-2022 redistricting cycle exists in a variety of formats across as many platforms, and not all states collect enough comment data for statistical inference. Many of the states that collect a sufficient sample of public comments, furthermore, do not ask for commenters to provide any information about their home city, their county, their ZIP Code, or otherwise. This information provides important context for our evaluation of how well commenters represent the demographic makeup of their communities. Of the states asking for geographic identifiers, Colorado has by far the largest sample of commenters, with over 5800 responses collected between March 12th, 2021 and October 11th, 2021. A Colorado Open Records Act (CORA) request was filed to collect an initial batch of comment data, and the returned responses were combined with comments collected by a web scraper spanning a web comment archive for Colorado's Independent Redistricting Commission. The vast majorty of comment responses include a first and a last name of the commenter as well as their ZIP Code of residence. Figure 1 illustrates the distribution of commenters by ZIP Code.

Comment lengths range from a measly 4 characters to 32,464 characters, a veritable publication in of itself. Commenters weighed in about any number of redistricting concerns, many of which did not correspond to a request about any specific geographic configuration. One Coloradoan comments two words, "Simply Hogwash", in July of 2021, while many others referenced an attached map or letter, and others still commented on issues like district competitiveness or incumbency protection. These comments do not lend much to measurements of congruence with any empirical map, but thankfully, the majority of comments make requests regarding communities of interest, usually referencing specific locations throughout the state. The comment below, submitted in late August of 2021, reflects several of the nuanced geographic configurations that commenters discuss, including requesting that locations be kept whole, grouped together, or separated from other locations.

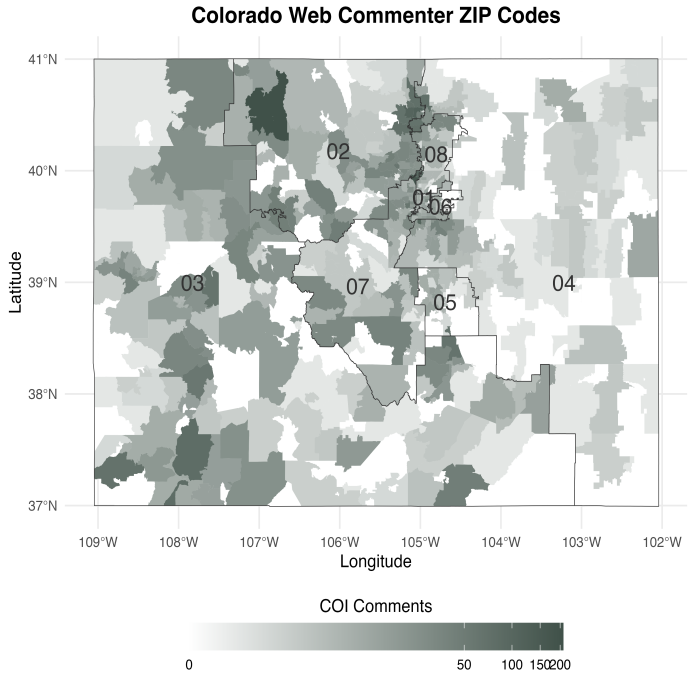


Figure 1: Collected Colorado web comments by ZIP Code with 2022 District boundaries overlaid

To Whom It May Concern:

I am very concerned about the preliminary State House Map presented to the public. The map as drawn makes little or no sense, discriminates against communities of color, and eliminates competitive districts, helping to demolish our free and fair democratic elections in El Paso County.

Why are former House Districts 17 and 18 being broken up? Both House Districts are communities of interest. House District 18 is already geographically connected from Manitou Springs to Old Colorado City to downtown Colorado Springs. This area is comprised of older homes with similar interests and needs. House District 17 is another community of interest, serving the needs of Southeastern Colorado Springs. Destroying House District 17 is a disservice to the communities of color that lie within its boundaries. Both of these districts are competitive. Changing them would nullify that fact and certainly make them non-competitive.

Furthermore, I am mystified by your delineation of the boundaries of the proposed House District 49. It comprises only half of the Tri-Lakes area, making absolutely no sense. The Tri-Lakes area is another community of interest and should be kept together as a geographically contiguous district. The Tri-Lakes area of Monument and Palmer Lake has totally different needs from Colorado Springs. Colorado Springs should definitely not be apportioned to the proposed House District 49.

I hope that in the future this State House Map contains competitive districts and accommodates communities of interest, not blatant political opportunism and gerrymandering. This is still a representative democracy, a republic, and public officials such as yourselves need to act in a forthright fashion and champion the democratic ideals of our republic.

Respectfully submitted,

Charlotte gagne

2.3 Annotating Public Web Comments

Figure 2 represents Charlotte's requests of the Colorado Independent Redistricting Commission as a network, with each distinct location as a node and edge representing pairwise relationships. Colorado's uniquely large and descriptive comment data makes it an ideal case study state for determining the distribution of commenter demographics as well as their networks of coordinated advocacy for particular requests. However, building networks of commenter requests requires that comments pass through three stages of annotation. The first stage assesses whether a comment discusses communities of interest and any specific locations pertaining to those communities. The second state involves extracting all named locations in a comment that pertain to a redistricting request. The third stage involves evaluating the relationships between these locations. From networks of relationships, not only can we identify clusters of commenters making similar requests across ZIP Codes and submission dates, but we can also measure the extent to which any given redistricting plan respects a particular request.

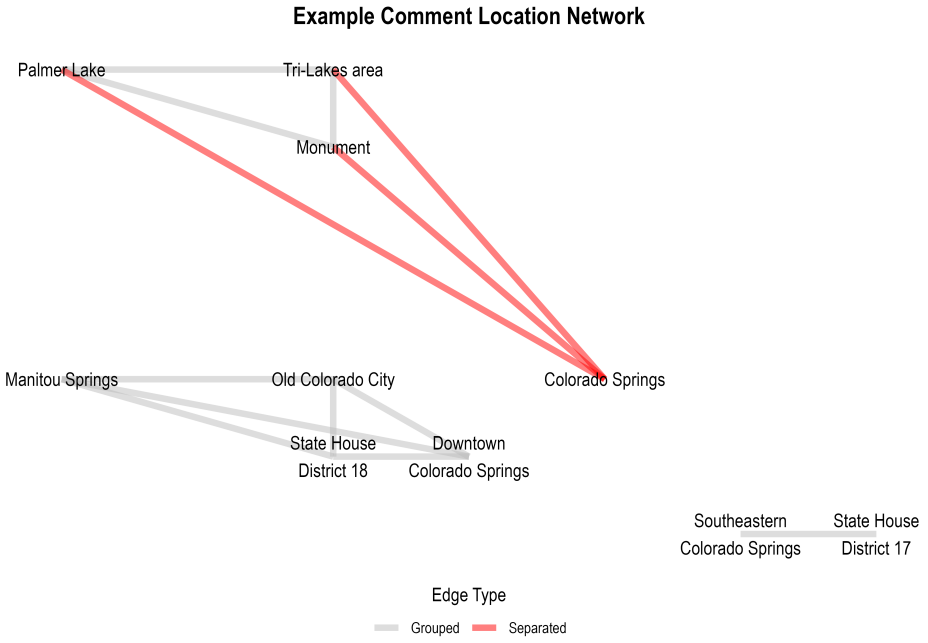


Figure 2: Example network of geographic configurations requested by the commenter across the Colorado Springs Metro Area

For each comment annotation stage, we look towards frontier methods of structured data extraction using

large language models (LLMs). LLM-based annotation has emerged as a defensible method of extracting information from political texts, with a variety of approaches suggested based on experiments with major models (Ornstein, Blasingame, and Truscott 2025; Benoit et al. 2026; Liu and Shi 2025; Heseltine and Clemm von Hohenberg 2024). Replicability, however, is of primary concern when using LLMs given their "black box" nature, which is why Barrie, Palmer, and Spirling advocate for the use of open source and open-weight models (2025). Our approach therefore makes use of `ellmer`, a recently-developed R package for structured data extraction from free text (Wickham et al. 2025), as well as Mistral AI's open source mistral-large-2512 model. The package enables the integration of engineered prompts with a complex output schema, allowing us to constrain location name and relationship responses for compatibility with downstream analyses. LLM annotations were compared against a ground truth set of 50 comments for Colorado that was annotated by the researchers, as well as a ground truth set for every other state in the cross-state analysis, in order to refine precision, recall, and overall accuracy. Details about prompting language and validation tests can be found in Appendix 5.1.

Across 4095 comments pertaining to communities of interest in Colorado, our location extraction process returned over 2000 unique locations and over 24,000 mentions overall. Request extraction returned almost 50,000 total requests, with almost 20,000 being unique. Many pairwise or single location requests were made by hundreds of commenters. Table 1 summarizes the 20 most popular extracted requests. In Colorado, commenters seem to have emphasized the distinction of large regions of the state as well as key towns in Northern Colorado like Fort Collins and Boulder.

Table 1: Popular requests made by Colorado commenters

Request	Commenters
Western Slope kept whole	336
Front Range separated from Western Slope	262
Fort Collins kept whole	244
Broomfield kept whole	207
Larimer County kept whole	201
Boulder County separated from Western Slope	186
Fort Collins grouped with Larimer County	178
Boulder County kept whole	158
Routt County kept whole	154
Boulder County grouped with Larimer County	149
Summit County kept whole	147
Boulder separated from Western Slope	145
Eastern Plains separated from Fort Collins	143
Eagle County kept whole	139
Rural Colorado kept whole	120
Larimer County separated from Weld County	113
Eagle County grouped with Routt County	111
Fort Collins grouped with Loveland	110
Fort Collins separated from Weld County	106
San Luis Valley kept whole	106

3 Results

4 Discussion

5 Appendix

5.1 Appendix A: Comment Data Extraction with Ellmer

The `ellmer` package in R was used to interact with off-the-shelf LLM chat features and extract structured data from public testimony comments (Wickham et al. 2025). Barrie, Palmer, and Spirling (2025) recommend the use of open-weight Large Language Models (LLMs), which allow for more consistent replication of coding and classification tasks compared to the more fragile but advanced proprietary frontier LLMs. Therefore, we make use of the "Mistral Large 2.0" model, available on OpenRouter (https://openrouter.ai/), combined with a three-stage `ellmer` schema to gather comment metadata, location information, and location relationships for comments deemed . Mistral Large 2.0 was chosen for its strengths in named entity recognition and structured data extraction from administrative texts (https://mistral.ai/industry/public-sector/).

5.1.1 Stage One - COI Relevance Screening

Many states offered the public a general portal through which to comment on any redistricting concern. Some commenters thus used these portals to voice frustration or approval about traditional redistricting outcomes such as compactness, competitiveness, incumbency protection, and minority representation, while others weighed in about procedural matters, transparency, and institutional reform. In order to screen for comments concerning communities of interest that mention specific locations to geocode, the following prompt was paired with each comment, and output was constrained to a binary structured data type with `ellmer`:

> Binary indicator measuring the specificity of this {{state}} commenter's description of a community of interest. A value of 1 indicates that the commenter is clearly discussing particular, named locations that they want kept together or separate for a district map. A value of 0 indicates that the commenter fails to mention specific locations or that they are discussing a different redistricting concern.

Only comments meeting COI relevance criteria were including in downstream data extraction steps. Because

prompting was consistent across all comments, screening was conducted for each state using a single parallel structured chat call.

5.1.2 Stage Two - Comment Metadata Extraction and Named Entity Recognition (NER)

The second stage of the `ellmer` pipeline was applied to every comment meeting COI relevance criteria. In this stage, a highly-structured prompting architecture was built to identify named geographic locations that were part of a specific redistricting configuration requested by the commenter. While other geographic NER software has proven successful at similar tasks (Halterman 2023), we chose to use `ellmer` for its flexibility in extracting more detailed location information as well as its use of well-maintained, open-source models and its compatibility with other R packages used throughout our analysis. Using a parallel structured chat with `ellmer`, the following system level prompt was used to first introduce the Mistral model to the task at hand:

> You are an expert geocoding research assistant evaluating public comments from the 2021-2022 redistricting cycle in the United States for quantitative downstream evaluation of institutional compliance with constituent input. The goal is to extract consistently named and formatted locations and comment-level metadata. The provided comments were all written by concerned citizens from {{localContext}}.

The `localContext` variable was based on any state, county, or ZIP Code level identifiers available within comment level metadata. For example, Colorado comment data included commenter ZIP Codes, so "ZIP Code XXXXX in Colorado" was assigned to `localContext` for all comments from the corresponding area. Each set of comments sharing the same value of `localContext` were bundled and processed with a separate parallel structured chat call. If location context was only available at the state level, prompts were bundled by temporal strata such as the date of the comment submission.

Each specific location mentioned throughout the comment was defined by a prompt schema of its own, all of which were embedded in an array type response from `ellmer`. This schema consisted of an enum (factor) response describing the location's administrative level, a string response for the location name, an enum

response for any cardinal direction subareas or other codifiable subareas relevant to the mentioned location, and a binary response reflecting whether the location was mentioned purely as context for an argument or narrative rather than being included in a requested configuration. The following top-level prompt for location recognition also included several rules designed to catch edge cases and constrain location mentions to those specifically mentioned by the commenter in a redistricting request:

> Return individual geographic locations, including landmarks, neighborhoods, townships, boroughs, towns, cities, school districts, counties, legislative districts, and regions, in accordance with the following rules:
>
> - Return locations that the commenter asks be kept together during redistricting.
> - Return locations that the commenter asks be grouped with another location during redistricting.
> - Return locations that the commenter asks be separated from another location during redistricting.
> - Do not return locations outside the commenter's state.
> - Do not return any proposed legislative districts discussed by the commenter, only existing districts as of 2020.
> - Do not return roads or interstates mentioned by the commenter.

Since proposed legislative districts have no history of representing a community, we asked the Mistral model to only include legislative districts that were in place prior to the 2021-2022 redistricting cycle. These existing districts could reasonably have corresponded to a territorial community at the time that public comments were collected.

The prompts for individual location schema components are shown below in order of appearance, including the prompts for a location's name, administrative level, descriptive subareas, and relevance to a redistricting request:

> Best estimate of the type or administrative level of the location. Must be exclusive and decisive.

If the location's administrative level is unclear, return 'NA'.

Official and vernacular administrative levels for mentioned locations vary from state to state. Therefore, the factor levels available for administrative level prompt outputs were selected from the following set based on each state's relevant properties, including the presence of congressional districts and town/township/borough/city naming conventions for municipalities: "Landmark", "School", "Neighborhood", "Township", "Borough", "Town", "City", "School District", "County", "State House District", "State Senate District", "Congressional District", "Region", and "NA".

After finding the location's administrative level, return the canonical, administrative name of the location. Unless the location is a region, do not include any subareas in the name (i.e. return 'Washington County' if the comment mentions 'Northern Washington County' or 'Rural Washington County'). Location names should thus read like the following rules:

Naming rules:

- Landmark: use the landmark's canonical name.
- Neighborhood: use the neighborhood's canonical name.
- Township: use the municipality's canonical name (do not include 'township' in the name).
- Borough: use the municipality's canonical name.
- Town: use the municipality's canonical name.
- City: use the municipality's canonical name.
- School District: use '[Name] School District'.
- County: use '[Name] County'.
- State House District: use 'State House District [Number]'.
- State Senate District: use 'State Senate District [Number]'.
- Congressional District: use 'Congressional District [Number]'.
- Region: Name must exactly match one of the permitted region names.

Permitted region names: {{stateRegions}}

The naming rules for the prompt above were designed to constrain outputs for each administrative level to the naming structure of Census-defined places available through the `tigris` package in R (Walker 2025). The `stateRegions` variable was defined based on statewide cardinal direction subareas (i.e. "Northern Colorado") as well as common vernacular regions in each state (i.e. "Front Range", "Happy Valley", or "Atlanta Metro Area"). Although these sets of regions are not exhaustive, our inclusion of this variable as a prompt rule rather than a separate structured prompt allows the Mistral model to deviate from the provided set and include alternative region names when no clear choice is available among `stateRegions`.

> A subarea specific to the location, if explicitly mentioned by the commenter. If the commenter refers to the entirety of the location (i.e. 'Washington County'), return 'NA'. Proper names of a location or region (i.e. 'Westridge', 'South Bend', 'Northern California') do not imply a separate subarea mention (i.e. 'Western', 'Southern', and 'Northern'). Return 'NA' in these cases.

Subarea options included the following: "Northern", "Northeastern", "Eastern", "Southeastern", "Southern", "Southwestern", "Western", "Northwestern", "Central", "Unincorporated", "Rural", "Suburban", "Urban", "Downtown", "NA".

> The relevance of the location to a specific request by the commenter regarding a community of interest and a possible legislative district boundary choice. Locations should be regarded as 'contextual' when they provide background information or anecdotes that are tangential to genuine geographic configurations and the broader thesis of the comment. Relevant locations are those that a commenter requests be kept whole, grouped with other locations, or separated from other locations. The following are examples of when 'contextual' should be returned for a mentioned location.
>
> - I was at the meeting in Springfield last night. Here are my observations.
> - We need our community better represented in [State Capital City] and Washington DC.
> - Our neighborhood has been split, as have many neighborhoods across Washington County.
> - I came back to this community after attending college at [State University].

- My comments address several concerns I have over the districts in the western part of the state.

This relevance prompt and its associated rules were added based on edge cases observed during the process of reading randomly sampled comments. Since "contextual" locations are not part of a genuine geographic configuration requested by the commenter, they should not be included in downstream analysis steps, especially including the evaluation of district plan congruence with constituent opinion.

5.1.3 Stage Three - Location Relationship Extraction

For any comments identified as having multiple locations mentioned, a third set of `ellmer` prompts was used to extract the relationships between locations according to the commenter's request. Individual relationships were embedded in an array type `ellmer` output. Since the nature of these relationships are dependent on a commenter's potentially complex and multifaceted arguments and their approval or disapproval of either existing or proposed district maps, the prompt for this classifier included several examples and stipulations, as shown below:

> This commenter is providing public input about a legislative redistricting cycle, requesting that some geographic locations they mention either be kept together or separated into various districts based on their communities of interest. Return any pairwise location relationships requested by the commenter. Interpret the commenter's request, complaint, or concern. Do not treat the composition of an existing or proposed district as a desired relationship unless the commenter explicitly endorses or rejects that composition.
>
> The following are examples of a 'grouped' pairwise relationship between 'Springfield' and 'Washington County':
>
> - 'Vote no on the proposed map, which separates Springfield from Washington County.'
> - 'Springfield and Washington County have been separated. This is a clear effort to Gerrymander our community.'

- ‘Springfield and Washington County share similar economic interests.’
- ‘Why is Springfield in a different district than Washington County?’
- ‘Separating Springfield from Washington County is ridiculous.’
- ‘Please Keep Springfield together with Washington County.’
- ‘Don’t split Springfield from Washington County.’
- ‘Springfield should be reunited with Washington County.’

The following are examples of a ‘separated’ pairwise relationship between ‘Springfield’ and ‘Washington County’:

- ‘Vote no on the proposed map, which combines Springfield with Washington County.’
- ‘Springfield and Washington County have been combined. This is a clear effort to Gerrymander our community.’
- ‘Springfield and Washington County do not share similar economic interests.’
- ‘Why is Springfield in the same district as Washington County?’
- ‘Combining Springfield and Washington County is ridiculous.’
- ‘Please Keep Springfield separated from Washington County.’
- ‘Don’t group Springfield with Washington County.’
- ‘Springfield should be kept apart from Washington County.’

The prompt schema for each relationship included a first location name response, a second location name response, a relationship type response, and a district type response. Location name responses, while not tied to individual prompt descriptions, were constrained to an enum output, with factor levels derived from the composite information for each location extracted in the previous step of the pipeline. The straightforward relationship type prompting was as follows:

> Return the nature of the relationship between the two locations

Relationship types were constrained to be either “grouped”, “separated”, or “unclear”. District type prompting consisted of the following:

> Return the legislative district type relevant to this pairwise relationship request, if mentioned by the commenter. If the relevant district types are unclear or if multiple types are relevant, return 'NA'.

District types relevant to the location relationship were constrained to be one or more of the following: "State House", "State Senate", "Congressional", or "NA" when no district type was listed or when multiple types were being discussed simultaneously.

5.1.4 Validation

For each state, 50 comments screened for the mention of communities of interest were randomly sampled and annotated by hand. These annotated comments followed the same rules as the prompting architecture described above, and from location mentions and requests among pairs of locations, a "Ground Truth" set of comments was constructed to validate `ellmer` output against. This ground truth set includes 592 mentioned locations and 329 groups of locations. For each validation test, two comparison sets of locations or requests were compared to a ground truth set using linear regression with robust standard errors under the following model:

$$Y = \beta_0 + State$$

Where Y is a precision or recall contrast between the two methods against the ground truth at the unit level, β_0 is an intercept term, and *State* is a state-wise fixed effect. In the context of our analysis, precision is the number of correctly identified units as a proportion of all identified units, while recall is the number of correctly identified units as a proportion of ground truth units. Regression coefficient tests below are against the null hypothesis that the intercept term is indistinguishable from 0, except in the cases of location level precision because ground truth and comparison unit sets are not necessarily of the same size, prohibiting the use of unit-wise contrast. In these cases, regression coefficient tests compare against a null hypothesis that a binary coefficient differentiating the two comparison methods is indistinguishable from 0.

5.1.4.1 Location Recognition Validation Table 2 explores the efficacy of the relevance field prompting. Removing contextual locations offered statistically significant increases to precision while only slightly diminishing recall, so this field was deemed useful for the purposes of consolidating downstream location request options.

Table 2: Unit contrast regression tests of full baseline location set (Method A) compared to a baseline location with contextual locations removed (Method B)

Test	Unit	Method A	Method B	Coefficient	P-Value
Location Recall	Location	0.863	0.834	-0.005	0.318
Location Recall	Comment	0.844	0.835	-0.007	0.319
Location Precision	Location	0.689	0.752	0.063	0.008
Location Precision	Comment	0.681	0.740	0.059	0.099

We also investigated precision and recall across multiple iterations of each location recognition task. Table 3 compares a method filtering mentioned locations that were mentioned at least twice across three iterations to locations mentioned in a single iteration. Table 4 compares a set for which mentions were mentioned in all three iterations to the same single iteration. Restricting locations to majority or unanimous mentions was associated with some small tradeoffs between precision and recall, with no definitive advantage to the task's broader accuracy, so we concluded that single-iteration extraction tasks were sufficient for downstream analysis.

Table 3: Unit contrast regression tests of single iteration locations (Method A) compared to locations mentioned across a majority of three iterations (Method B)

Test	Unit	Method A	Method B	Coefficient	P-Value
Location Recall	Location	0.834	0.841	0.014	0.257
Location Recall	Comment	0.835	0.839	0.013	0.128
Location Precision	Location	0.752	0.756	0.004	0.869
Location Precision	Comment	0.740	0.753	0.013	0.706

Table 4: Unit contrast regression tests of single iteration locations (Method A) compared to locations mentioned across unanimously three iterations (Method B)

Test	Unit	Method A	Method B	Coefficient	P-Value
Location Recall	Location	0.834	0.814	-0.023	0.058
Location Recall	Comment	0.835	0.826	-0.009	0.389
Location Precision	Location	0.752	0.797	0.045	0.055
Location Precision	Comment	0.740	0.797	0.057	0.108

The baseline prompting architecture, as described above, was developed iteratively, succeeding several other variants as the idiosyncrasies of the `ellmer` package and its interactions with the Mistral model were discovered. Table 5 below shows a comparison between the baseline prompting architecture for location recognition above and a simpler prompting architecture lacking rules at the system prompt and contextual location identification levels. This simpler architecture was not statistically distinguishable from the more verbose architecture, but the intercept term differences for the latter were signed in the direction of higher precision and recall, leading us to designate it as our baseline.

Table 5: Unit contrast regression tests of baseline prompting architecture (Method A) compared to a simpler prompting architecture with fewer prompt examples (Method B)

Test	Unit	Method A	Method B	Coefficient	P-Value
Location Recall	Location	0.846	0.826	-0.042	0.060
Location Recall	Comment	0.847	0.816	-0.056	0.037
Location Precision	Location	0.760	0.775	0.015	0.535
Location Precision	Comment	0.761	0.768	0.007	0.848

Lastly, Table 6 compares a simpler prompting architecture with and without a system prompt. Although not statistically significant, substantively large increases to precision under the inclusion of a system prompt drove us to retain the prompt in our baseline architecture.

Table 6: Unit contrast regression tests of simpler prompting architecture without a system prompt (Method A) compared to a similar prompting architecture with a system prompt (Method B)

Test	Unit	Method A	Method B	Coefficient	P-Value
Location Recall	Location	0.878	0.853	0.014	0.468
Location Recall	Comment	0.843	0.840	0.006	0.838
Location Precision	Location	0.680	0.722	0.045	0.059
Location Precision	Comment	0.679	0.713	0.034	0.340

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